

计算机学院面向对象技术与方法期末试卷样卷

【一般选择题 15 道题 30 分, 填空/写运行结果 4 道题 40 分, 编程 2 道题 30 分】

班级_____ 学号_____ 姓名_____ 成绩_____

(注意: 把所有答案写在答题纸上; 交卷时把试卷与答题纸分开交。试卷反面可以做草稿)

Part I. (30 points) Choose A, B, C or D and write your choices on the Answer Sheet.

1. _____ can not be initialized in the initialization list of the constructor.
(A) members of user-defined classes (B) const data members
(C) static data members (D) references
2. _____ is right.
(A) const int *p; p = new int(8); (B) int * const p; p = new int(8);
(C) const int &p; p = new int(8); (D) int & const p; p = new int(8);
3. After executing the following codes, _____ is right.
#include <iostream>
using namespace std;
void main()
{
int a(66), b(55), *c = &a;
int* &p = c;
p = &b;
cout << a << ", " << b << ", " << *c << ", " << *p << endl;
}
(A) 66, 55, 55, 55 (B) 55, 66, 66, 66
(C) 66, 55, 66, 55 (D) 55, 66, 55, 66

4.

5.

6.

.....

Part II. (40 points) Fill blanks or write outputs.

1. (10 points) Fill blanks and write outputs.

```
class Sample
{
public:
    Sample()    { value = 15; }
    int GetValue(_____ obj) _____
    {    return obj.value + 18; }
private:
    int value;
```

```
};
void main( )
{
    const Sample S;
    S.GetValue( S);
}
```

2. (10 points) Write outputs.

```
#include <iostream>
using namespace std;
class Sample
{
public:
    Sample (int x = 10 ) : a(x)  { c = a + 50;  cout << c << endl; }
    ~Sample ( )  { c = c - 30;  cout << c << endl; }
    static int GetC( )  { return c; }
private:
    static int c;
    int a;
};
int Sample::c = 40;
void main( )
{
    Sample c1, c2(15), c3(25);
    cout << c2.GetC( ) << endl;
}
```

3.

4.

PartIII. (30 points) Programming.

1. (15 points) Define a class **Fraction** (分数), which has two data members: **numerator**(分子) and **denominator**(分母), initialized in the constructor. Users can use **Fraction** in **main()** as follows:

```
int main( )
{
    Fraction R1(1, 2),  R2(2, 3),  R3;
    R3 = R1 + R2;
    double number = R3;
    cout << R3 << endl;
    cout << number << endl;
    return 0;
}
```

2. (15 points)